Chapter 1

Probabilistic learning: A Summary

- Observed data (features and targets) obtained by randomly sampling:
  - $\mathcal{X}$ according to probability distribution $p_M(\mathbf{x})$.
  - $\mathcal{Y}$ according to conditional distribution $p_C(t|\mathbf{x})$.
- Probabilistic predictor: derive from training set $\mathcal{T}$ a conditional distribution $p^*(t|\mathbf{x})$ approximating the unknown p_C.
- A **decision strategy** is then applied to return a specific prediction $h(\mathbf{x})$, typically:

$$h(\mathbf{x}) = \underset{t \in \mathcal{Y}}{\operatorname{argmax}}\ p^*(t|\mathbf{x}) \tag{1.1}$$

Decision theory

- Goal: choose a decision rule that minimizes **expected cost** (loss).

Decision Regions and Classification

- Partition input space $\mathcal{X}$ into K disjoint **decision regions** $\mathcal{R}_1, \dots, \mathcal{R}_K$.
- Point $\mathbf{x}$ in $\mathcal{R}_k$ assigned to class C_k.
- For binary classification $\{C_0, C_1\}$, probability of correct classification:

$$\begin{aligned}\hat{p}(\mathbf{x}) &= p(\mathbf{x} \in \mathcal{R}_0, C_0) + p(\mathbf{x} \in \mathcal{R}_1, C_1) \\ &= \int_{\mathcal{R}_0} p(\mathbf{x}, C_0) d\mathbf{x} + \int_{\mathcal{R}_1} p(\mathbf{x}, C_1) d\mathbf{x}\end{aligned}$$

- To maximize, assign $\mathbf{x}$ to C_0 if $p(\mathbf{x}, C_0) > p(\mathbf{x}, C_1)$.
- Using product rule $p(\mathbf{x}, C_k) = p(C_k|\mathbf{x})p(\mathbf{x})$, condition becomes:

$$\frac{p(C_0|\mathbf{x})}{p(C_1|\mathbf{x})} > 1 \tag{1.2}$$

- Optimal strategy: choose class with highest **posterior probability** $p(C_k|\mathbf{x})$.
- For $K > 2$ classes:

$$h(\mathbf{x}) = \underset{0 \le i \le K-1}{\operatorname{argmax}}\ p(C_i|\mathbf{x}) \tag{1.3}$$

Incorporating Misclassification Costs

- **Loss matrix** L_{ij}: cost of classifying item as $\mathcal{C}_j$ when it actually belongs to $\mathcal{C}_i$.
- Expected cost (risk) for decision regions:

$$\mathbb{E}[L] = \sum_i \sum_j \int_{\mathcal{R}_j} L_{ij} p(\mathbf{x}, \mathcal{C}_i) d\mathbf{x} = \sum_j \int_{\mathcal{R}_j} \left[\sum_i L_{ij} p(\mathcal{C}_i|\mathbf{x}) \right] p(\mathbf{x}) d\mathbf{x} \tag{1.4}$$

- To minimize, for each $\mathbf{x}$, choose class k minimizing **conditional risk**:

$$j = \underset{k}{\operatorname{argmin}} \sum_i L_{ik} p(\mathcal{C}_i|\mathbf{x}) \tag{1.5}$$

- **Binary case with costs:**
  - Cost for $\mathcal{C}_0$: $L_{00} p(\mathcal{C}_0|\mathbf{x}) + L_{10} p(\mathcal{C}_1|\mathbf{x})$.
  - Cost for $\mathcal{C}_1$: $L_{01} p(\mathcal{C}_0|\mathbf{x}) + L_{11} p(\mathcal{C}_1|\mathbf{x})$.
  - Assign $\mathbf{x}$ to $\mathcal{R}_0$ if:

$$L_{00} p(\mathcal{C}_0|\mathbf{x}) + L_{10} p(\mathcal{C}_1|\mathbf{x}) < L_{01} p(\mathcal{C}_0|\mathbf{x}) + L_{11} p(\mathcal{C}_1|\mathbf{x})$$
$$\frac{p(\mathcal{C}_0|\mathbf{x})}{p(\mathcal{C}_1|\mathbf{x})} > \frac{L_{01} - L_{11}}{L_{10} - L_{00}}$$

  - With threshold $\theta = \frac{L_{01} - L_{11}}{L_{10} + L_{01} - L_{00} - L_{11}}$, assign to $\mathcal{C}_0$ if $p(\mathcal{C}_0|\mathbf{x}) > \theta$.
  - If errors have equal cost and correct predictions zero cost, $\theta = 1/2$.

Medical Diagnosis: $\mathcal{C}_0$ = Sick, $\mathcal{C}_1$ = Healthy. Loss matrix:

$$L = \begin{pmatrix} 0 & 100 \\ 1 & 0 \end{pmatrix} \tag{1.6}$$

Missing a sick patient ($L_{01} = 100$) costs 100 times more than false alarm ($L_{10} = 1$). Assignment rule for "Healthy":

$$p(\mathcal{C}_1|\mathbf{x}) > \frac{L_{01}}{L_{10} + L_{01}} = \frac{100}{101} \approx 0.99$$

Only classify as Healthy with 99% confidence.

The Reject Option

- Introduce **rejection region** when uncertainty is too high.

- In binary case, classify as C_0 only if expected cost significantly lower by factor $(1-\epsilon)$:

$$h(\mathbf{x}) = \begin{cases} 0 & \text{if } p(C_0|\mathbf{x}) > \theta' \\ 1 & \text{if } p(C_0|\mathbf{x}) < 1-\theta'' \\ \text{undefined} & \text{otherwise} \end{cases} \tag{1.7}$$

- θ' and $1-\theta''$ derived from comparing costs scaled by safety margin $(1-\epsilon)$.
- If misclassification costs higher than correct classification costs, $\theta' > \theta$ (harder to commit without strong evidence).

Decision Theory for Regression

- For continuous target t, common loss is **squared loss** $L(t, y(\mathbf{x})) = (y(\mathbf{x}) - t)^2$.
- Goal: find $h(\mathbf{x})$ minimizing expected loss:

$$\mathbb{E}[L] = \int \int (h(\mathbf{x}) - t)^2 p(\mathbf{x}, t) d\mathbf{x} dt \tag{1.8}$$

- Let $\mathbb{E}[t|\mathbf{x}] = \int t p(t|\mathbf{x}) dt$ be conditional mean.
- Decompose squared error:

$$\begin{aligned} (h(\mathbf{x}) - t)^2 &= (h(\mathbf{x}) - \mathbb{E}[t|\mathbf{x}] + \mathbb{E}[t|\mathbf{x}] - t)^2 \\ &= (h(\mathbf{x}) - \mathbb{E}[t|\mathbf{x}])^2 + (\mathbb{E}[t|\mathbf{x}] - t)^2 + 2(h(\mathbf{x}) - \mathbb{E}[t|\mathbf{x}])(\mathbb{E}[t|\mathbf{x}] - t) \end{aligned}$$

- Taking expectation over t for fixed $\mathbf{x}$, last term vanishes.
- Total expected loss:

$$\mathbb{E}[L] = \int (h(\mathbf{x}) - \mathbb{E}[t|\mathbf{x}])^2 p(\mathbf{x}) d\mathbf{x} + \int \int (\mathbb{E}[t|\mathbf{x}] - t)^2 p(\mathbf{x}, t) d\mathbf{x} dt \tag{1.9}$$

- Second term independent of $h(\mathbf{x})$ (intrinsic noise).
- To minimize first term, set it to zero $\Rightarrow$ optimal prediction:

$$h^*(\mathbf{x}) = \mathbb{E}[t|\mathbf{x}] \tag{1.10}$$

Approximating the Conditional Distribution

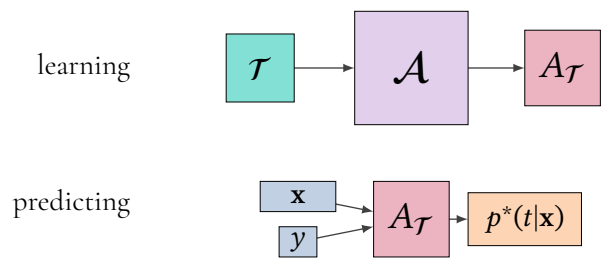
- True $p_C(t|\mathbf{x})$ unknown; we have only finite training set $\mathcal{T}$.
- Goal: infer approximation $p(t|\mathbf{x})$ as close as possible to p_C.

Generative vs. Discriminative Approaches

1. **Generative Probabilistic Models**: Model class-conditional densities $p(\mathbf{x}|\mathcal{C}_k)$ and priors $p(\mathcal{C}_k)$. Obtain posterior via Bayes' rule:
$$p(\mathcal{C}_k|\mathbf{x}) = \frac{p(\mathbf{x}|\mathcal{C}_k)p(\mathcal{C}_k)}{p(\mathbf{x})} = \frac{p(\mathbf{x}|\mathcal{C}_k)p(\mathcal{C}_k)}{\sum_j p(\mathbf{x}|\mathcal{C}_j)p(\mathcal{C}_j)}$$
For comparing classes, only need $p(\mathbf{x}|\mathcal{C}_k)p(\mathcal{C}_k)$.
2. **Discriminative Probabilistic Models**: Directly estimate posterior class probabilities $p(\mathcal{C}_k|\mathbf{x})$ from training set.

Three-Step Learning Procedure

1. **Model Selection**: Define class of possible conditional distributions $\mathcal{P}$ (hypothesis space).
2. **Inference**: Select "best" distribution $p^* \in \mathcal{P}$ based on training data using quality measure q (e.g., likelihood).
3. **Prediction**: For new $\mathbf{x}$, use $p^*(t|\mathbf{x})$ to assign probabilities.



Parametric Distributions and the Logistic Example

- **Parametric approach**: distributions share mathematical structure governed by parameters $\mathbf{w}$.
- Learning becomes search for parameters that best fit data.

Logistic Regression for Binary Classification:

- Target $t \in \{0, 1\}$ follows Bernoulli distribution conditioned on input:
$$p(t|\mathbf{x}, \mathbf{w}) = \pi(\mathbf{x})^t(1 - \pi(\mathbf{x}))^{1-t} \tag{1.11}$$
- $\pi(\mathbf{x}) = p(t = 1|\mathbf{x})$ modeled via logistic sigmoid:
$$\pi(\mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x}) = \frac{1}{1 + e^{-(\sum_{i=1}^d w_i x_i + w_0)}} \tag{1.12}$$
- Squashes linear output into $[0, 1]$ for probability interpretation.

Probabilistic models

Parametric and Non-Parametric Probabilistic Models

- **Probabilistic model**: structured collection of probability distributions over a data domain.
- Instances identified by **parameters** $\boldsymbol{\theta}$.

Bivariate Gaussian model: Determined by:

1. Mean vector $\boldsymbol{\mu} = (\mu_1, \mu_2)$ (center).
2. Covariance matrix $\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}$ (spread and orientation, with $\sigma_{12} = \sigma_{21}$).

Parametric Models: Fixed, finite parameter set predefined before seeing data.
Non-Parametric Models: Number of parameters grows with dataset size, adapting complexity to data.

The Concept of Likelihood

- **Likelihood** $L(\boldsymbol{\theta}|\mathcal{T})$: probability of observing dataset $\mathcal{T}$ given parameters $\boldsymbol{\theta}$:

$$L(\boldsymbol{\theta}|\mathcal{T}) = p(\mathcal{T}|\boldsymbol{\theta}) \tag{1.13}$$

- Conceptual difference: probability (fixed $\boldsymbol{\theta}$, data varies) vs likelihood (fixed data $\mathcal{T}$, vary $\boldsymbol{\theta}$).
- Frequentist interpretation: parameters are fixed, deterministic, unknown constants.
- Under i.i.d. assumption, joint probability factorizes.
- **Density Estimation** (only inputs $\mathbf{X}$):

$$L(\boldsymbol{\theta}|\mathcal{T}) = p(\mathbf{X}|\boldsymbol{\theta}) = \prod_{i=1}^{n} p(\mathbf{x}_i|\boldsymbol{\theta}) \tag{1.14}$$

- **Supervised Learning** (input-target pairs $(\mathbf{X}, \mathbf{t})$):

$$\begin{aligned} L(\boldsymbol{\theta}|\mathcal{T}) = p(\mathbf{X}, \mathbf{t}|\boldsymbol{\theta}) &= \prod_{i=1}^{n} p(\mathbf{x}_i, t_i|\boldsymbol{\theta}) = \prod_{i=1}^{n} p(t_i|\mathbf{x}_i, \boldsymbol{\theta}) p(\mathbf{x}_i|\boldsymbol{\theta}) \\ &= p(\mathbf{X}|\boldsymbol{\theta}) \prod_{i=1}^{n} p(t_i|\mathbf{x}_i, \boldsymbol{\theta}) \end{aligned}$$

- If $p(\mathbf{x}|\boldsymbol{\theta})$ uniform or independent of $\boldsymbol{\theta}$, likelihood proportional to product of conditionals:

$$L(\boldsymbol{\theta}|\mathcal{T}) \propto \prod_{i=1}^{n} p(t_i|\mathbf{x}_i, \boldsymbol{\theta}) \tag{1.15}$$

Maximum Likelihood Estimation (MLE)

- MLE principle: choose parameters $\boldsymbol{\theta}^*$ that make observed data most probable:

$$\boldsymbol{\theta}^*_{ML} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\, L(\boldsymbol{\theta}|\mathcal{T}) \tag{1.16}$$

- In practice, use **log-likelihood** to avoid numerical underflow:

$$l(\boldsymbol{\theta}|\mathcal{T}) = \ln L(\boldsymbol{\theta}|\mathcal{T}) \tag{1.17}$$

- Maximizing l equivalent to maximizing L (log is monotonic).
- Products become sums:

$$\boldsymbol{\theta}^*_{ML} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \sum_{i=1}^{n} \ln p(\mathbf{x}_i|\boldsymbol{\theta}) \quad \text{or} \quad \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \sum_{i=1}^{n} \ln p(t_i|\mathbf{x}_i, \boldsymbol{\theta}) \tag{1.18}$$

- Stationary point condition:

$$\nabla l(\boldsymbol{\theta}|\mathcal{T}) = \mathbf{0} \tag{1.19}$$

  (necessary but not sufficient; verify via Hessian).
- Once $\boldsymbol{\theta}^*_{ML}$ computed:

$$p(\mathbf{x}|\mathbf{X}) \approx p(\mathbf{x}|\boldsymbol{\theta}^*_{ML})$$
$$p(t|\mathbf{x}, \mathbf{X}, \mathbf{t}) \approx p(t|\mathbf{x}, \boldsymbol{\theta}^*_{ML})$$

Bernoulli Trials: For n binary events with success probability ϕ:

- Likelihood: $L(\phi|\mathbf{X}) = \phi^{n_1}(1-\phi)^{n_0}$ (n_1 successes, n_0 failures).
- Log-likelihood: $l(\phi|\mathbf{X}) = n_1 \ln \phi + n_0 \ln(1-\phi)$.
- Setting derivative to zero: $\frac{n_1}{\phi} - \frac{n_0}{1-\phi} = 0 \implies \phi^*_{ML} = \frac{n_1}{n}$.
- Intuitive frequentist result: observed sample frequency.

Linear Regression with Gaussian Noise:

- Model: $p(t|\mathbf{x}, \mathbf{w}, \sigma^2) = \mathcal{N}(\mathbf{w}^T\mathbf{x} + w_0, 1/\beta)$ with precision β.
- Likelihood: $L(\mathbf{t}|\mathbf{X}, \mathbf{w}, w_0, \beta) = \prod_{i=1}^{n} \mathcal{N}(\mathbf{w}^T\mathbf{x}_i + w_0, \beta)$.
- Log-likelihood:

$$l(\mathbf{t}|\mathbf{X}, \mathbf{w}, w_0, \beta) = -\frac{\beta}{2}\sum_{i=1}^{n}(\mathbf{w}^T\mathbf{x}_i + w_0 - t_i)^2 + \frac{n}{2}\ln\beta - \frac{n}{2}\ln(2\pi)$$

- Partial derivatives:

$$\frac{\partial}{\partial w_k} l = -\frac{\beta}{2} \sum_{i=1}^{n} (\mathbf{w}^T \mathbf{x}_i + w_0 - t_i) x_{ik}$$
$$\frac{\partial}{\partial w_0} l = -\frac{\beta}{2} \sum_{i=1}^{n} (\mathbf{w}^T \mathbf{x}_i + w_0 - t_i)$$
$$\frac{\partial}{\partial \beta} l = -\frac{1}{2} \sum_{i=1}^{n} (\mathbf{w}^T \mathbf{x}_i + w_0 - t_i)^2 + \frac{n}{2\beta}$$

- ML estimates for $\mathbf{w}, w_0$: solution of $(d+1, d+1)$ linear system:

$$\sum_{i=1}^{n} (\mathbf{w}^T \mathbf{x}_i + w_0 - t_i) x_{ik} = 0 \quad k = 1, \dots, d$$
$$\sum_{i=1}^{n} (\mathbf{w}^T \mathbf{x}_i + w_0 - t_i) = 0$$

- ML estimate for β:

$$\beta_{ML} = \left(\frac{1}{n} \sum_{i=1}^{n} (\mathbf{w}^T \mathbf{x}_i + w_0 - t_i)^2 \right)^{-1} \tag{1.20}$$

Overfitting and the Maximum a Posteriori (MAP) Estimate

- MLE drawback: **overfitting** (memorizing noise, poor generalization).
- Solution: add **penalty function** $P(\boldsymbol{\theta})$ to log-likelihood:

$$C(\boldsymbol{\theta}|\mathbf{X}) = l(\boldsymbol{\theta}|\mathbf{X}) - P(\boldsymbol{\theta}) \tag{1.21}$$

- Common choice: weight decay $P(\boldsymbol{\theta}) = \frac{\gamma}{2} \|\boldsymbol{\theta}\|^2$.
- **Maximum a Posteriori (MAP)**: treat $\boldsymbol{\theta}$ as random variable with **prior distribution** $p(\boldsymbol{\theta})$.
- Using Bayes' theorem:

$$p(\boldsymbol{\theta}|\mathcal{T}) = \frac{p(\mathcal{T}|\boldsymbol{\theta}) p(\boldsymbol{\theta})}{p(\mathcal{T})} \propto L(\boldsymbol{\theta}|\mathcal{T}) p(\boldsymbol{\theta}) \tag{1.22}$$

- MAP estimate maximizes posterior:

$$\boldsymbol{\theta}^*_{MAP} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \; (\ln L(\boldsymbol{\theta}|\mathcal{T}) + \ln p(\boldsymbol{\theta})) \tag{1.23}$$

- For Gaussian prior $p(\boldsymbol{\theta}) \sim \mathcal{N}(\boldsymbol{\theta}|\mathbf{0}, \sigma^2)$:

$$\begin{aligned}\boldsymbol{\theta}^*_{MAP} &= \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \ (l(\boldsymbol{\theta}|\mathcal{T}) + \ln p(\boldsymbol{\theta})) \\ &= \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \left(l(\boldsymbol{\theta}|\mathcal{T}) - \frac{\|\boldsymbol{\theta}\|^2}{2\sigma^2} \right)\end{aligned}$$

- This equals penalty function with $\gamma = \frac{1}{\sigma^2}$. Regularization = prior belief that parameters should not be too large.

MAP for Bernoulli with Beta Prior:

- Prior: Beta distribution $p(\phi|\alpha, \beta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \phi^{\alpha-1}(1-\phi)^{\beta-1}$.
- Log-likelihood: $l(\phi|\mathbf{X}) = n_1 \ln \phi + n_0 \ln(1-\phi)$.
- MAP estimation: set derivative of $l(\phi|\mathbf{X}) + \ln p(\phi|\alpha, \beta)$ to zero:

$$\phi^*_{MAP} = \frac{n_1 + \alpha - 1}{n_0 + n_1 + \alpha + \beta - 2} = \frac{n_1 + \alpha - 1}{n + \alpha + \beta - 2} \tag{1.24}$$

- Prevents estimate from becoming 0 or 1 when data is scarce ("black swan" problem).

Full Bayesian Inference

- Posterior distribution:

$$p(\boldsymbol{\theta}|\mathbf{X}) = \frac{p(\mathbf{X}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{X})} = \frac{p(\mathbf{X}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{\int_{\boldsymbol{\theta}} p(\mathbf{X}|\boldsymbol{\theta})d\boldsymbol{\theta}} \tag{1.25}$$

- MAP gives mode; may be inaccurate for skewed or multimodal distributions.
- **Fully Bayesian** approach considers entire posterior distribution.
- **Posterior mean** (expectation):

$$\boldsymbol{\theta}^* = \mathbb{E}[\boldsymbol{\theta}|\mathbf{X}] = \int_{\boldsymbol{\theta}} \boldsymbol{\theta} p(\boldsymbol{\theta}|\mathbf{X}) d\boldsymbol{\theta} \tag{1.26}$$

Bernoulli with Beta Prior (Bayesian):

- Prior: $p(\phi|\alpha, \beta) = \text{Beta}(\phi|\alpha, \beta)$.
- Posterior:

$$\begin{aligned}p(\phi|\mathbf{X}, \alpha, \beta) &= \frac{\phi^{n_1}(1-\phi)^{n_0}\phi^{\alpha-1}(1-\phi)^{\beta-1}}{Z} \\ &= \frac{\phi^{n_1+\alpha-1}(1-\phi)^{n_0+\beta-1}}{Z}\end{aligned}$$

- Hence $p(\phi|\mathbf{X}, \alpha, \beta) = \text{Beta}(\phi|\alpha + n_1, \beta + n_0)$.

Model Selection

- Dilemma: before estimating parameters $\boldsymbol{\theta}$, must choose **model structure** (hypothesis space): functional form, features, hyper-parameters.
- **Model selection**: identify among candidates $\{m_1, \dots, m_r\}$ the one expected to best represent underlying process.
- Not looking for best fit to training data, but model whose *best instantiation* generalizes to unseen data.
- Underfitting: too simple model fails to capture essential patterns.
- Overfitting: too complex model captures random noise.

Validation Strategies

Test Set Validation: Partition dataset into **Training set** (learn parameters) and **Test set** (held out for unbiased evaluation). Efficient for large datasets; problematic for small datasets.

Cross-Validation: K-fold CV: partition into K equal subsets. In K iterations, one subset as test, union of remaining $K-1$ as training. Final performance = average across folds.

Leave-One-Out (LOO) Cross-Validation: Special case $K = n$ (one point left out each iteration). Maximizes training data but extremely time-consuming.

Information-Theoretic Measures

- Provide score based on training performance adjusted by "complexity penalty."
- Balance **goodness-of-fit** and **parsimony**.

Akaike Information Criterion (AIC): Rooted in information theory (KL divergence). Estimates relative information lost when using a model.

$$AIC = 2|\boldsymbol{\theta}| - 2\log p(\mathbf{X}|\boldsymbol{\theta}_{ML}) = 2|\boldsymbol{\theta}| - 2\max \boldsymbol{\theta} l(\boldsymbol{\theta}|\mathbf{X}) \tag{1.27}$$

Bayesian Information Criterion (BIC): Derived from Bayesian model evidence. Assumes one candidate is "true" model.

$$\begin{aligned} BIC &= |\boldsymbol{\theta}| \log n - 2\log p(\mathbf{X}|\boldsymbol{\theta}_{ML}) \\ &= |\boldsymbol{\theta}| \log |\mathbf{X}| - 2\max \boldsymbol{\theta} l(\boldsymbol{\theta}|\mathbf{X}) \end{aligned}$$

Interpreting AIC vs BIC

- AIC penalty: constant 2 per parameter.
- BIC penalty: $\log n$ per parameter (grows with dataset size).
- As n increases, BIC penalizes complexity much more heavily.

- **Philosophy:**
  - AIC: minimize prediction error (favors slightly more complex models if they improve fit).
  - BIC: consistent (if true model among candidates, BIC selects it as $n \to \infty$).
- In practice: AIC for prediction, BIC for explanation/identifying "correct" structure.

Polynomial Model Selection:

- For dataset $n = 100$ with true quadratic relationship:

Degree (M)	$\lvert\theta\rvert$	Deviance ($-2 \ln L$)	AIC	BIC
1 (Linear)	2	150.0	154.0	159.2
2 (Quadratic)	**3**	**120.0**	**126.0**	**133.8**
3 (Cubic)	4	119.5	127.5	137.9
4 (Quartic)	5	119.2	129.2	142.2

- Both AIC and BIC identify quadratic as best.
- For AIC, cubic is close second (difference 1.5).
- For BIC, quadratic clearly separated from more complex alternatives.

Bayesian Model Comparison and Averaging

Model Averaging and Mixture Predictive Distribution

- Bayesian approach treats model selection as accounting for uncertainty.
- Consider L candidate models $\{\mathcal{M}_i\}_{i=1}^{L}$.
- Prior belief: $p(\mathcal{M}_i)$.
- Posterior after observing $\mathcal{T} = (\mathbf{X}, \mathbf{t})$:

$$p(\mathcal{M}_i|\mathcal{T}) = \frac{p(\mathcal{T}|\mathcal{M}_i)p(\mathcal{M}_i)}{p(\mathcal{T})} \propto p(\mathcal{T}|\mathcal{M}_i)p(\mathcal{M}_i) \tag{1.28}$$

- $p(\mathcal{T}|\mathcal{M}_i)$: **marginal likelihood** or **model evidence**.
- **Bayes factor** comparing models i and j:

$$K = \frac{p(\mathcal{T}|\mathcal{M}_i)}{p(\mathcal{T}|\mathcal{M}_j)} \tag{1.29}$$

- **Model Averaging** predictive distribution:

$$p(t|\mathbf{x}, \mathcal{T}) = \sum_{i=1}^{L} p(t|\mathbf{x}, \mathcal{M}_i, \mathcal{T})p(\mathcal{M}_i|\mathcal{T}) \tag{1.30}$$

- Final prediction = weighted average of individual model predictions, weights = posterior probabilities.

The Anatomy of Model Evidence

- Marginal likelihood: average of likelihood $p(\mathcal{T}|\mathbf{w}, \mathcal{M}_i)$ over all parameter values, weighted by prior:

$$p(\mathcal{T}|\mathcal{M}_i) = \int p(\mathcal{T}|\mathbf{w}, \mathcal{M}_i)p(\mathbf{w}|\mathcal{M}_i)d\mathbf{w} \tag{1.31}$$

- Interpretation: probability of generating $\mathcal{T}$ by first sampling $\mathbf{w}$ from prior, then sampling data from model.
- Evidence is also normalization constant for posterior of parameters:

$$p(\mathbf{w}|\mathcal{T}, \mathcal{M}_i) = \frac{p(\mathcal{T}|\mathbf{w}, \mathcal{M}_i)p(\mathbf{w}|\mathcal{M}_i)}{p(\mathcal{T}|\mathcal{M}_i)} \tag{1.32}$$

Insight into Complexity: Occam's Razor

- Consider model with single parameter w.
- Assume posterior sharply peaked around w_{MAP} with width Δw_{pos}.
- Assume prior uniform with width Δw_{pri}, so $p(w|\mathcal{M}) = 1/\Delta w_{pri}$.
- Approximate evidence integral:

$$\begin{aligned} p(\mathcal{T}|\mathcal{M}) &= \int p(\mathcal{T}|w, \mathcal{M})p(w|\mathcal{M})dw \\ &\simeq p(\mathcal{T}|w_{MAP}, \mathcal{M})\frac{\Delta w_{pos}}{\Delta w_{pri}} \end{aligned}$$

- Log evidence:

$$\log p(\mathcal{T}|\mathcal{M}) \simeq \log p(\mathcal{T}|w_{MAP}, \mathcal{M}) + \log\left(\frac{\Delta w_{pos}}{\Delta w_{pri}}\right) \tag{1.33}$$

- **Data fit**: $\log p(\mathcal{T}|w_{MAP}, \mathcal{M})$ (favors complex models).
- **Complexity penalty (Occam factor)**: second term, negative since $\Delta w_{pos} < \Delta w_{pri}$.
- Small Δw_{pos} (sensitive model) $\Rightarrow$ large negative penalty.
- For M independent parameters:

$$\log p(\mathcal{T}|\mathcal{M}) \simeq \log p(\mathcal{T}|\mathbf{w}_{MAP}, \mathcal{M}) + M\log\left(\frac{\Delta w_{pos}}{\Delta w_{pri}}\right) \tag{1.34}$$

- Bayesian evidence naturally embodies **Occam's Razor**: prefers simplest explanation that fits data.
- Interpretation:
  - Simple model: restricted, can only describe small range of datasets; if data outside range, evidence near zero.

- Complex model: can fit vast array of datasets, but spreads probability mass thin, so evidence for any specific dataset small.
- Intermediate model: flexible enough to capture structure but not so flexible that probability mass is too diluted ⇒ highest evidence.

Model Evidence for Linear Regression:

- Likelihood:

$$p(\mathbf{t}|\mathbf{X}, \mathbf{w}, \beta) = \left(\frac{\beta}{2\pi}\right)^{n/2} \exp\left(-\frac{\beta}{2}\|\mathbf{t} - \mathbf{X}\mathbf{w}\|^2\right) \tag{1.35}$$

- Gaussian prior with precision α:

$$p(\mathbf{w}|\alpha) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \alpha^{-1}\mathbf{I}) = \left(\frac{\alpha}{2\pi}\right)^{M/2} \exp\left(-\frac{\alpha}{2}\mathbf{w}^T\mathbf{w}\right) \tag{1.36}$$

- Model evidence:

$$p(\mathbf{t}|\mathbf{X}, \alpha, \beta) = \int p(\mathbf{t}|\mathbf{X}, \mathbf{w}, \beta)p(\mathbf{w}|\alpha)d\mathbf{w} \tag{1.37}$$

- Combine exponents: $\mathbb{E}[\mathbf{w}] = \frac{\beta}{2}\|\mathbf{t} - \mathbf{X}\mathbf{w}\|^2 + \frac{\alpha}{2}\mathbf{w}^T\mathbf{w}$
- Complete square: $\mathbb{E}[\mathbf{w}] = \mathbb{E}[\mathbf{w}_{MAP}] + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{MAP})^T\mathbf{A}(\mathbf{w} - \mathbf{w}_{MAP})$
- $\mathbf{A} = \alpha\mathbf{I} + \beta\mathbf{X}^T\mathbf{X}$ (Hessian), $w_{MAP} = \beta\mathbf{A}^{-1}\mathbf{X}^T\mathbf{t}$.
- Integral evaluates to $(2\pi)^{M/2}|\mathbf{A}|^{-1/2}$.
- Log-evidence:

$$\log p(\mathbf{t}|\mathbf{X}, \alpha, \beta) = \frac{M}{2}\log\alpha + \frac{n}{2}\log\beta - \mathbb{E}[\mathbf{w}_{MAP}] - \frac{1}{2}\log|\mathbf{A}| - \frac{n}{2}\log(2\pi) \tag{1.38}$$

- **Interpretation:**
  - Data fit: $-\mathbb{E}[\mathbf{w}_{MAP}]$ (sum of squared errors).
  - Occam penalty: $-\frac{1}{2}\log|\mathbf{A}| + \frac{M}{2}\log\alpha$ (log of ratio of posterior-to-prior uncertainty).

Evidence Framework for Hyper-parameter Tuning

- Maximize model evidence with respect to hyper-parameters α, β.
- **Optimal α:**
  - Let λ_i be eigenvalues of $\beta\mathbf{X}^T\mathbf{X}$, so eigenvalues of $\mathbf{A}$ are $(\lambda_i + \alpha)$.
  - Define $\gamma = \sum_{i=1}^{M} \frac{\lambda_i}{\lambda_i + \alpha}$ (number of "well-determined parameters").
  - Stationary condition yields iterative update:

$$\alpha = \frac{\gamma}{\mathbf{w}_{MAP}^T\mathbf{w}_{MAP}} \tag{1.39}$$

- **Optimal β:**

$$\frac{1}{\beta} = \frac{1}{n-\gamma} \sum_{i=1}^{n} \{t_i - \mathbf{w}_{MAP}^T \mathbf{x}_i\}^2 \tag{1.40}$$

- γ interpretation ($0 \leq \gamma \leq M$):
  - If $\lambda_i \gg \alpha$: parameter well-determined by data, contribution to $\gamma \approx 1$.
  - If $\lambda_i \ll \alpha$: prior dominates, parameter shrunk toward zero, contribution to $\gamma \approx 0$.
- Framework performs **internal validation**: measures information from data vs prior, adjusts regularization (α) and noise level (β) accordingly.

Summary of Bayesian Model Selection

- **Automatic Occam's Razor**: complex models penalized by posterior-to-prior volume ratio.
- **No need for test set**: hyper-parameters tuned by maximizing evidence on training data alone.
- **Quantified uncertainty**: predictions as mixture, acknowledging model uncertainty.
- Cohesive, theoretically grounded approach to building models that generalize effectively while providing realistic uncertainty measures.